

4.1.1 离散型变量的数学期望

平均数: 180, 190, 200 $\rightarrow \frac{180+190+200}{3}$

加权平均: 90, 100, 15

90% 9% 1% [权]

$$0.9 \times 90 + 0.09 \times 100 + 0.01 \times 15$$

离散型随机变量如果 $P(X=x_k)=P_k$

若 $\sum_{k=1}^{\infty} x_k P_k$ 绝对收敛, $E_x = \sum_{k=1}^{\infty} x_k P_k$

*: 数学期望并非都存在.

定义中要求级数 $\sum_{k=1}^{\infty} x_k P_k$ 绝对收敛,

保证了级数的和与求和顺序无关.

例: 甲、乙两人生产产品, x_1, x_2 分别代表
两人生产的次品数

x_1	0	1	2	3
P	0.3	0.3	0.2	0.2

甲的技术水平高.

$$\begin{aligned} \text{解: } EX_1 &= 0 \times 0.3 + 1 \times 0.3 + 2 \times 0.2 + 3 \times 0.2 \\ &= 1.3 \end{aligned}$$

$$EX_2 = 0.5 + 0.6 = 1.1 < 1.3$$

$\therefore$ 乙的水平高.

例: 一等品 二等品 三等品 次品
概率 60% 20% 10% 10%
产值 6元 4.8元 4元 0元
求平均产值.

$$\begin{aligned} E_x &= 6 \times 0.6 + 4.8 \times 0.2 + 4 \times 0.1 + 0 \times 0.1 \\ &= 4.96 \text{ 元} \end{aligned}$$

4.1.2 连续型的期望

随机变量 X , 它的密度函数为 $f(x)$

若 $\int_{-\infty}^{+\infty} x f(x) dx$ 绝对收敛, 则

$$E_x = \int_{-\infty}^{+\infty} x f(x) dx.$$

*: 连续型的数学期望并非都存在.

$x \rightarrow$ 取值 $f(x) \rightarrow$ 概率

$$\text{例: } f(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & \text{o.w} \end{cases} \text{ 求 } E_x$$

$$\begin{aligned} E_x &= \int_{-\infty}^{+\infty} x f(x) dx \\ &= \int_0^1 x \cdot 2x dx \\ &= \frac{2}{3} \end{aligned}$$

例: 家电先使用, 后付款.

电器寿命服从 $\lambda = \frac{1}{10}$ 的指数分布.

$X \leq 1$ 年 1500元

$1 < X \leq 2$ 年 2000元

$2 < X \leq 3$ 年 2500元

$X \geq 3$ 年 3000元

收费 Y 的期望

$$\begin{aligned} \text{解: } P(X \leq 1) &= \int_0^1 \frac{1}{10} e^{-\frac{x}{10}} dx \\ &= 0.0952 \end{aligned}$$

$$\begin{aligned} P(1 < X \leq 2) &= \int_1^2 \frac{1}{10} e^{-\frac{x}{10}} dx \\ &= 0.0861 \end{aligned}$$

$$P(2 < X \leq 3) = 0.0779$$

$$P(X > 3) = 0.7408.$$

Y	1500	2000	2500	3000
P	0.0952	0.0861	0.0779	0.7408

$$\begin{aligned} EY &= 1500 \times 0.0952 + 2000 \times 0.0861 \\ &\quad + 2500 \times 0.0779 + 3000 \times 0.7408 = 2732.15 \end{aligned}$$



扫描全能王 创建

4.1.3 随机变量函数的期望

$$EX \xrightarrow{Y=g(X)} EY$$

$$\text{离散 } \sum x_i p_i \quad \left| \quad \sum g(x_i) p_i\right.$$

$$\text{连续 } \int_{-\infty}^{+\infty} x f(x) dx \quad \left| \quad \int_{-\infty}^{+\infty} g(x) f(x) dx\right.$$

例:

X	0	1	2
P	0.1	0.6	0.3

 $Y = 4X + 1$

求Y的期望

解:

Y	1	5	9
P	0.1	0.6	0.3

$$EY = 1 \times 0.1 + 5 \times 0.6 + 9 \times 0.3 = 5.8$$

也可直接根据X算Y.

$$EY = (4 \times 0 + 1) \times 0.1 + (4 \times 1 + 1) \times 0.6 + (4 \times 2 + 1) \times 0.3 = 5.8$$

$$\text{法} \because EX = 0 \times 0.1 + 1 \times 0.6 + 2 \times 0.3 = 1.2$$

$$\begin{aligned} E(X - EX)^2 &= E(X - 1.2)^2 \\ &= (0 - 1.2)^2 \times 0.1 + (1 - 1.2)^2 \times 0.6 + (2 - 1.2)^2 \times 0.3 \\ &= 0.36 \end{aligned}$$

例: $f(x) = \begin{cases} \frac{1}{2} & 0 \leq x \leq 2 \\ 0 & \text{o.w.} \end{cases}$ 求 $Y = 4X + 1$ 的期望.

$$\begin{aligned} EY &= E(4X + 1) = \int_{-\infty}^{+\infty} (4x + 1) f(x) dx \\ &= \int_0^2 \frac{1}{2} (4x + 1) dx = 5. \end{aligned}$$

例: X: 商品需求量, 为 $[2000, 4000]$ 上的均匀分布. 每卖一吨赚3万, 卖不出则每吨损失1万.

解: 设y为准备出口数量, Y为收益.

$$Y = g(x) = \begin{cases} 3y & x \geq y \\ 3x - (y - x) & x < y \end{cases}$$

$$f(x) = \begin{cases} \frac{1}{2000} & 2000 \leq x \leq 4000 \\ 0 & \text{o.w.} \end{cases}$$

$$\begin{aligned} EY &= \int_{-\infty}^{+\infty} g(x) f(x) dx \\ &= \int_{2000}^{4000} \frac{1}{2000} g(x) dx \\ &= \int_{2000}^y \frac{1}{2000} (4x - y) + \int_y^{4000} \frac{1}{2000} 3y dx \\ &= \frac{1}{1000} (-y^2 + 7000y - 4 \times 10^6) \end{aligned}$$

y=3500 时Y最大.

二维变量函数. $Z = g(X, Y)$ 的期望.

(X, Y)

$$1) \text{ 离散型 } EZ = \sum_i \sum_j g(x_i, y_j) \cdot f(x_i, y_j) P_{ij}$$

X \ Y	0	1	2
1	0.1	0.1	0.2
2	0.2	0.2	0.2

$$\text{求 } Z = X^2 - Y$$

$$\begin{aligned} EZ &= (1^2 - 0) \times 0.1 + (1^2 - 1) \times 0.1 + (1^2 - 2) \times 0.2 \\ &\quad + (2^2 - 0) \times 0.2 + (2^2 - 1) \times 0.2 + (2^2 - 2) \times 0.2 \end{aligned}$$

$$2) \text{ 连续型: } EZ = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} g(x, y) f(x, y) dx dy$$

例: $f(x, y) = \begin{cases} x + y & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{o.w.} \end{cases}$

$$\begin{aligned} E(XY) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy f(x, y) dx dy \\ &= \int_0^1 \int_0^1 xy (x + y) dx dy \\ &= \frac{1}{3} \end{aligned}$$



例: X : 进货量 Y : 需求量 两者相互独立
且均满足 $[10, 20]$ 上的均匀分布
卖一件获利 1000 元. 其他商店调剂
一件获利 500 元, 求平均利润.

解: $Z = g(X, Y) = \begin{cases} 1000Y & Y \leq X \\ 1000X + 500(Y-X) & Y > X \end{cases}$

$$f_X(x) = \begin{cases} \frac{1}{10} & 10 \leq x \leq 20 \\ 0 & \text{o.w.} \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{1}{10} & 10 \leq y \leq 20 \\ 0 & \text{o.w.} \end{cases}$$

$$f(x, y) = f_X(x) f_Y(y) = \begin{cases} \frac{1}{100} & 10 \leq x \leq 20, 10 \leq y \leq 20 \\ 0 & \text{o.w.} \end{cases}$$

$$\begin{aligned} E_Z &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} g(x, y) f(x, y) dx dy \\ &= \frac{1}{100} \int_{10}^{20} \int_{10}^{20} g(x, y) dx dy \\ &= \frac{1}{100} \int_{10}^{20} dy \int_{10}^{20} g(x, y) dx \\ &= \frac{1}{100} \int_{10}^{20} dy \left[\int_{10}^y 500(x+y) dx + \int_y^{20} 1000y dx \right] \\ &= 14166.67 \end{aligned}$$

4.1.4 数学期望的性质

(1) $E(C) = C$

(2) $E(X+C) = EX + C$ $E(X+C) = \int_{-\infty}^{+\infty} (x+c) f(x) dx$

(3) $E(CX) = CEX$ $= \int_{-\infty}^{+\infty} x f(x) dx + C \int_{-\infty}^{+\infty} f(x) dx$

(4) $E(kX+b) = kEX + b$ $= EX + C$

(5) $E(X \pm Y) = EX \pm EY$ (任可时候均成立)

$$\begin{aligned} E(X \pm Y) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} (x \pm y) f(x, y) dx dy \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x f(x, y) dx dy \pm \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} y f(x, y) dx dy \\ &= \int_{-\infty}^{+\infty} x dx \int_{-\infty}^{+\infty} f(x, y) dy \pm \int_{-\infty}^{+\infty} y dy \int_{-\infty}^{+\infty} f(x, y) dx \\ &= \int_{-\infty}^{+\infty} x f_X(x) dx \pm \int_{-\infty}^{+\infty} y f_Y(y) dy = EX \pm EY \end{aligned}$$

推论:

$$E(\sum C_i X_i) = \sum C_i EX_i$$

$$E(\frac{1}{n} \sum X_i) = \frac{1}{n} \sum EX_i$$

例: $E(2X_1 - 3X_2 + 4X_3)$
 $= 2EX_1 - 3EX_2 + 4EX_3$

(6) X, Y 独立时有

$$E(XY) = EX \cdot EY$$

$$\begin{aligned} \text{证: } E(XY) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy f(x, y) dx dy \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy f_X(x) f_Y(y) dx dy \\ &= \int_{-\infty}^{+\infty} x f_X(x) dx \int_{-\infty}^{+\infty} y f_Y(y) dy \\ &= \int_{-\infty}^{+\infty} x f_X(x) dx \cdot \int_{-\infty}^{+\infty} y f_Y(y) dy \\ &= EX \cdot EY \end{aligned}$$

能用性质尽量用性质

例: X, Y 相互独立. 求 $E(X+Y), E(XY), EY^2$

X	9	10	11
P	0.3	0.5	0.2

Y	6	7
P	0.4	0.6

求 $E(X+Y), E(XY), EY^2$

1) $E(X+Y) = EX + EY$

$$\begin{aligned} &= 9 \times 0.3 + 10 \times 0.5 + 11 \times 0.2 + 6 \times 0.4 + 7 \times 0.6 \\ &= 9.9 + 6.6 \\ &= 16.5 \end{aligned}$$

2) $E(XY) = EX \cdot EY$

$$\begin{aligned} &= 9.9 \times 6.6 \\ &= 65.34 \end{aligned}$$

3) $E(Y^2) = 36 \times 0.4 + 49 \times 0.6$
 $= 43.8$



例

X \ Y	1	2	3	
0	0.1	0.2	0.3	0.6
1	0.2	0.1	0.1	0.4
	0.3	0.3	0.4	

求 $E(X-Y)$ 和 $E(XY)$

X	0	1
P	0.6	0.4

$$EX = 0.4$$

Y	1	2	3
P	0.3	0.3	0.4

$$EY = 0.3 + 2 \times 0.3 + 3 \times 0.4 = 2.1$$

$$1) E(X-Y) = E(X) - E(Y) = -1.7$$

2) $E(XY)$ 先验独立 $\rightarrow X, Y$ 不独立.

$$E(XY) = 1 \times 0.2 + 2 \times 0.1 + 3 \times 0.1 = 0.7$$

例: 扔 10 个骰子. X 表示点数之和.

求 EX .

解: 设 X_i 表示第 i 个骰子的点数.

X_i	1	2	3	4	5	6
P	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$EX_i = 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + \dots + 6 \times \frac{1}{6} = \frac{7}{2}$$

$$EX = E(X_1 + X_2 + \dots + X_{10})$$

$$= \sum_{i=1}^{10} EX_i$$

$$= 10 \times \frac{7}{2}$$

$$= 35.$$

例: N 件产品. M 件次品. 从中任取 n 件 ($n \leq M \leq N$). 求次品数的期望.

解: 次品: $1 \sim M$. 令 $X_i = \begin{cases} 1 & \text{第 } i \text{ 件次品被取出.} \\ 0 & \text{第 } i \text{ 件次品未被取出.} \end{cases}$
($i = 1, 2, \dots, M$)

X 表示次品数.

$$X = X_1 + X_2 + \dots + X_M$$

$$P\{X_i = 1\} = \frac{n}{N} \quad P\{X_i = 0\} = 1 - \frac{n}{N}$$

$$EX_i = \frac{n}{N} \times 1 + (1 - \frac{n}{N}) \times 0 = \frac{n}{N}$$

$$EX = EX_1 + EX_2 + \dots + EX_M$$

$$= \frac{nM}{N}$$

4.1.5 条件期望

一个变量取定某个值另一个变量的期望.

1. 离散型. $E(X|Y=y_j) = \sum x_i P(X=x_i|Y=y_j)$

$$E(Y|X=x_i) = \sum y_j P(Y=y_j|X=x_i)$$

X \ Y	0	1	2	3
0	0.1	0.2	0.3	
1	0.2	0.1	0.1	0.4

求 $E(Y|X=1)$

Y	1	2	3
$P(Y X=1)$	0.5	0.25	0.25

$$E(Y|X=1) = 1 \times 0.5 + 2 \times 0.25 + 3 \times 0.25 = 1.75.$$

2. 连续型

$$E(X|Y=y) = \int_{-\infty}^{+\infty} x f(x|y) dx$$

$$E(Y|X=x) = \int_{-\infty}^{+\infty} y f(y|x) dx.$$

▲: 条件期望具有期望的一切性质



4.2.1 方差

与期望的偏离程度 $(X-EX)^2$

方差: $DX = E(X-EX)^2$ 标准差: $\sqrt{DX}$

(1) 离散型: $DX = \sum_k (x_k - EX)^2 P_k$

(2) 连续型: $DX = \int_{-\infty}^{+\infty} (x - EX)^2 f(x) dx$

$$DX = EX^2 - (EX)^2$$

例: $\begin{array}{c|ccc} x & -2 & 0 & 2 \\ \hline p & 0.4 & 0.3 & 0.3 \end{array}$ 求 DX

$$EX = -2 \times 0.4 + 2 \times 0.3 = -0.2$$

$$EX^2 = 4 \times 0.4 + 4 \times 0.3 = 2.8$$

$$DX = EX^2 - (EX)^2 = 2.76$$

例: $f(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & o.w. \end{cases}$ 求 DX

$$\text{解: } EX = \int_0^1 x f(x) dx = \frac{2}{3}$$

$$EX^2 = \int_0^1 x^2 f(x) dx = \frac{1}{2}$$

$$DX = EX^2 - (EX)^2 = \frac{1}{18}$$

例: $f(x) = \begin{cases} ax^2 + bx + c & 0 < x < 1 \\ 0 & o.w. \end{cases}$

$$EX = 0.5, \quad DX = 0.15$$

$$\text{解: } \begin{cases} \int_0^1 (ax^2 + bx + c) dx = 1 \\ EX = \int_0^1 x(ax^2 + bx + c) dx = 0.5 \\ DX = \int_0^1 x^2(ax^2 + bx + c) dx - 0.5^2 = 0.15 \end{cases}$$

$$\begin{cases} \frac{a}{4} + \frac{b}{3} + \frac{c}{2} = 0.5 \\ \frac{a}{5} + \frac{b}{4} + \frac{c}{3} = 0.4 \\ \frac{a}{3} + \frac{b}{2} + c = 1 \end{cases}$$

$$\bar{A} = \begin{pmatrix} 3 & 4 & 6 & 6 \\ 12 & 15 & 20 & 24 \\ 2 & 3 & 6 & 6 \end{pmatrix} \xrightarrow{\text{行}}$$

4.2.2 方差的性质

$$(1) DC = 0 \quad DC = EC^2 - (EC)^2 = C^2 - C^2 = 0$$

$$(2) D(X+C) = DX$$

$$(3) D(CX) = C^2 DX$$

$$(4) D(KX+b) = K^2 DX$$

(5) X, Y 独立.

$$D(X \pm Y) = DX + DY$$

$$D(X \pm Y) = E(X \pm Y - E(X \pm Y))^2$$

$$= E(X \pm Y - EX \mp EY)^2$$

$$= E((X-EX) \pm (Y-EY))^2$$

$$= E[(X-EX)^2 + (Y-EY)^2 \pm 2(X-EX)(Y-EY)]$$

$$= E(X-EX)^2 + E(Y-EY)^2 \pm 2E[(X-EX)(Y-EY)]$$

$$E[(X-EX)(Y-EY)]$$

$$= E(XY - XEY - YEX + EXEY) \quad \text{常数}$$

$$= EX \cdot EY - EY \cdot EX - EX \cdot EY + EXEY$$

$$= 0$$

$$\therefore D(X \pm Y) = E(X-EX)^2 + E(Y-EY)^2$$

$$= DX + DY$$



推论: $D(X_1 + X_2 + \dots + X_n)$

$$= DX_1 + DX_2 + \dots + DX_n$$

$$(6) DX=0 \Leftrightarrow P(X=EX)=1$$

$$X^* = \frac{X-EX}{\sqrt{DX}} \quad EX^*=0 \quad DX^*=1$$

$$EX^* = E\left(\frac{X-EX}{\sqrt{DX}}\right) = \frac{1}{\sqrt{DX}}(EX-EX)=0$$

$$DX^* = \frac{1}{DX} D(X-EX) = \frac{1}{DX} DX = 1$$

4.3.1 常见离散型随机变量的期望与方差.

0-1分布: $P(X=k) = p^k(1-p)^{1-k} \quad k=0,1.$

X	0	1
P	$1-p$	p

$$EX = p \quad DX = EX^2 - (EX)^2 = p - p^2 = p(1-p) = pq.$$

二项分布: $P(X=k) = C_n^k p^k q^{n-k} \quad k=0,1,\dots,n$

$$EX = \sum_{k=0}^n k C_n^k p^k q^{n-k} = \sum_{k=0}^n k \cdot \frac{n!}{k!(n-k)!} p^k q^{n-k}$$

*: $k=0$ 时
该项为0

$$= \sum_{k=1}^n \frac{n!}{(k-1)!(n-k)!} p^k q^{n-k} = np \sum_{k=1}^n \frac{(n-1)!}{(k-1)!(n-k)!} p^{k-1} q^{n-k}$$

$$* : (p+q)^n = \sum_{k=0}^n C_n^k p^k q^{n-k}$$

$$\therefore EX = np \sum_{k=0}^{n-1} C_{n-1}^{k-1} p^{k-1} q^{n-k} = np (p+q)^{n-1} = np.$$

$$EX^2 = npq + n^2 p^2 \quad DX = npq.$$

X : 项分布总次数 $X_1 \dots X_n$
0/1 0/1.

$$X = X_1 + \dots + X_n$$

$$EX = E(X_1 + X_2 + \dots + X_n)$$

$$= EX_1 + \dots + EX_n$$

$$= np.$$

$$DX = D(X_1 + \dots + X_n)$$

$$= DX_1 + DX_2 + \dots + DX_n$$

$$= npq.$$

几何分布: $P(X=k)$ 首次发生.

$$P(X=k) = (1-p)^{k-1} \cdot p \quad k=1,2,3,\dots$$

$$\text{引理: } \sum_{k=1}^{\infty} k x^{k-1} = \left(\sum_{k=1}^{\infty} x^k \right)' = \left(\frac{x}{1-x} \right)' = \frac{1}{(1-x)^2} \quad |x| < 1.$$

$$EX = \sum_{k=1}^{\infty} k (1-p)^{k-1} p = p \cdot \left(\sum_{k=1}^{\infty} k (1-p)^{k-1} \right) = p \cdot \frac{1}{[1-(1-p)]^2} = \frac{1}{p}.$$

$$EX^2 = \sum_{k=1}^{\infty} k^2 (1-p)^{k-1} p$$

引理

$$\sum_{k=1}^{\infty} k^2 x^{k-1} = \sum_{k=1}^{\infty} k \cdot k x^{k-1}$$

$$= \left(\sum_{k=1}^{\infty} k x^k \right)'$$

$$= \left(x \sum_{k=1}^{\infty} k x^{k-1} \right)'$$

$$= \left(\frac{x}{(1-x)^2} \right)' = \frac{1+x}{(1-x)^3} \quad |x| < 1.$$



$$EX^2 = \frac{2-P}{P^2}$$

$$\therefore DX = \frac{1-P}{P^2}$$

泊松分布.

$$P(X=k) = \frac{\lambda^k}{k!} e^{-\lambda} \quad k=0,1,2,\dots$$

$$EX = \sum_{k=0}^{\infty} k \frac{\lambda^k}{k!} e^{-\lambda}$$

$$= \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!} e^{-\lambda}$$

$$= \lambda \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} e^{-\lambda}$$

$$= \lambda \sum_{m=0}^{\infty} \frac{\lambda^m}{m!} e^{-\lambda}$$

$$= \lambda \quad \text{泊松分布}$$

$$EX^2 = \sum_{k=0}^{\infty} k^2 \frac{\lambda^k}{k!} e^{-\lambda}$$

$$= \sum_{k=0}^{\infty} k \frac{\lambda^k}{(k-1)!} e^{-\lambda}$$

$$= \sum_{k=1}^{\infty} k \frac{\lambda^k}{(k-1)!} e^{-\lambda}$$

$$= \sum_{k=1}^{\infty} \frac{(k-1)\lambda^k}{(k-1)!} e^{-\lambda} + \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!} e^{-\lambda}$$

$$= \sum_{k=2}^{\infty} \frac{\lambda^k}{(k-2)!} e^{-\lambda} + \lambda \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} e^{-\lambda}$$

$$= \lambda^2 \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} e^{-\lambda} + \lambda$$

$$= \lambda^2 + \lambda$$

$$DX = EX^2 - (EX)^2 = \lambda$$

4.3.2 常见连续型的期望与方差

(1) 均匀分布

$$f(x) = \begin{cases} \frac{1}{b-a} & x \in [a, b] \\ 0 & \text{else} \end{cases}$$

$$EX = \int_a^b x \cdot \frac{1}{b-a} dx = \frac{a+b}{2}$$

$$EX^2 = \int_a^b x^2 \cdot \frac{1}{b-a} dx = \frac{b^2+ab+a^2}{3}$$

$$DX = EX^2 - (EX)^2 = \frac{(b-a)^2}{12}$$

(2) 指数分布

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & \text{else} \end{cases}$$

$$EX = \int_0^{+\infty} x \cdot \lambda e^{-\lambda x} dx$$

$$= - \int_0^{+\infty} x de^{-\lambda x}$$

$$= -x e^{-\lambda x} \Big|_0^{+\infty} + \int_0^{+\infty} e^{-\lambda x} dx$$

$$= \frac{1}{\lambda}$$

$$EX^2 = \int_0^{+\infty} x^2 \lambda e^{-\lambda x} dx$$

$$= - \int_0^{+\infty} x^2 de^{-\lambda x}$$

$$= -x^2 e^{-\lambda x} \Big|_0^{+\infty} + 2 \int_0^{+\infty} x e^{-\lambda x} dx$$

$$= \frac{2}{\lambda^2}$$

$$DX = EX^2 - (EX)^2 = \frac{1}{\lambda^2}$$

(3) 正态分布. $X \sim N(\mu, \sigma^2)$
 $\downarrow \quad \downarrow$
 $EX \quad DX$

$$Y = \frac{X-\mu}{\sigma} \sim N(0,1)$$

$$EY=0 \quad DY=1$$

$$X = \sigma Y + \mu \quad EX = E(\sigma Y + \mu)$$

$$= \sigma EY + \mu$$

$$= \mu$$

$$DX = D(\sigma Y + \mu) = \sigma^2 DY = \sigma^2$$



总结:

分布	定义	EX	DX
0-1	$P(X=k) = p^k(1-p)^{1-k} (k=0,1)$	p	pq
二项	$P(X=k) = C_n^k p^k q^{n-k} (k=0,1,\dots,n)$	np	npq
几何	$P(X=k) = (1-p)^{k-1} p (k=1,2,\dots)$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
泊松	$P(X=k) = \frac{\lambda^k}{k!} e^{-\lambda} (k=0,1,\dots)$	λ	λ
均匀	$f(x) = \begin{cases} \frac{1}{b-a} & x \in [a,b] \\ 0 & \text{else} \end{cases}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
指数	$f(x) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & \text{else} \end{cases}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
正态	$X \sim N(\mu, \sigma^2)$	μ	σ^2

4.4.1 独立方差

$$\begin{aligned}
 E[(X-EX)(Y-EY)] &= \text{Cov}(X, Y) \\
 &= E(XY - XEY - YEX + EXEY) \\
 &= E(XY) - EYEX - EXEY + EXEY \\
 &= E(XY) - EXEY
 \end{aligned}$$

$$D(X \pm Y) = DX + DY \quad (\text{独立})$$

$$= DX + DY \pm 2\text{Cov}(X, Y)$$

例:

X \ Y	-1	0	1	
-1	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{3}{8}$
0	$\frac{1}{8}$	0	$\frac{1}{8}$	$\frac{1}{4}$
1	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{3}{8}$
	$\frac{3}{8}$	$\frac{1}{4}$	$\frac{3}{8}$	

$$EX = -1 \times \frac{3}{8} + 1 \times \frac{3}{8} = 0$$

$$EY = 0$$

$$E(XY) = \frac{1}{8} - \frac{1}{8} - \frac{1}{8} + \frac{1}{8} = 0$$

求 $\text{Cov}(X, Y)$

Cov 性质

$$1) \text{Cov}(X, Y) = \text{Cov}(Y, X)$$

$$\text{左} = E(XY) - EXEY \quad \text{右} = E(YX) - EYEX$$

$$2) \text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$$

$$\text{左} = E(aXbY) - E(aX)E(bY)$$

$$= abE(XY) - abE(X)E(Y)$$

$$3) \text{Cov}(X_1 + X_2, Y) = \text{Cov}(X_1, Y) + \text{Cov}(X_2, Y)$$

$$\text{左} = E[(X_1 + X_2)Y] - E(X_1 + X_2)EY$$

$$= E(X_1Y) + E(X_2Y) - EX_1EY - EX_2EY$$

$$= \text{Cov}(X_1, Y) + \text{Cov}(X_2, Y)$$

$$\therefore \text{Cov}(XY) = 0 - 0 = 0. \text{ 不独立.}$$

$$f(x, y) = \begin{cases} x+y & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{else} \end{cases}$$

求 $\text{Cov}(X, Y)$

$$\text{解: } f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy \quad 0 \leq x \leq 1.$$

$$= \int_0^1 (x+y) dy$$

$$= xy + \frac{1}{2}y^2 \Big|_0^1$$

$$= x + \frac{1}{2}$$

$$\therefore f_X(x) = \begin{cases} x + \frac{1}{2} & 0 \leq x \leq 1 \\ 0 & \text{else} \end{cases}$$

$$f_Y(y) = \begin{cases} y + \frac{1}{2} & 0 \leq y \leq 1 \\ 0 & \text{else} \end{cases}$$

$$EX = \int_0^1 x(x + \frac{1}{2}) dx = \frac{7}{12}$$

$$EY = \frac{7}{12}$$

$$E(XY) = \int_0^1 \int_0^1 xy(x+y) dx dy = \frac{1}{3}$$

$$\text{Cov}(XY) = E(XY) - EXEY = -\frac{1}{14}$$



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4) $COV(C, X) = 0$

$$\begin{aligned} \text{左} &= E(CX) - E C E X \\ &= C E X - C E X \\ &= 0 \end{aligned}$$

5) X, Y 独立, $COV(X, Y) = 0$

$$E(XY) = E X E Y$$

$$COV(X, Y) = E(XY) - E X E Y$$

$COV(X, Y) \rightarrow$ 关系

受计量单位影响

$$X^* = \frac{X - EX}{\sqrt{DX}} \quad Y^* = \frac{Y - EY}{\sqrt{DY}} \quad \text{标准化}$$

$$COV(X^*, Y^*) = E(X^* Y^*) = EX^* \cdot EY^*$$

$$= E\left[\frac{X-EX}{\sqrt{DX}} \frac{Y-EY}{\sqrt{DY}}\right] - E\left(\frac{X-EX}{\sqrt{DX}}\right) E\left(\frac{Y-EY}{\sqrt{DY}}\right)$$

$$= E\left[\frac{X-EX}{\sqrt{DX}} \frac{Y-EY}{\sqrt{DY}}\right]$$

$$= \frac{E[(X-EX)(Y-EY)]}{\sqrt{DX} \sqrt{DY}}$$

$$= \frac{COV(X, Y)}{\sqrt{DX} \sqrt{DY}} = \rho$$

4.4.2 相关系数

$$\rho = \frac{COV(X, Y)}{\sqrt{DX} \sqrt{DY}} = \frac{E(XY) - EX EY}{\sqrt{DX} \sqrt{DY}}$$

与 $COV(X, Y)$ 同正, 同负, 同 0.

例: $DX=4 \quad DY=9 \quad \rho=0.5 \quad D(X-Y)$

$$D(X-Y) = DX + DY - 2COV(X, Y)$$

$$COV(X, Y) = \rho \sqrt{DX} \sqrt{DY} = 0.5 \times 2 \times 3 = 3$$

$$D(X-Y) = 4 + 9 - 2 \times 3 = 7$$

$$\therefore |\rho| \leq 1$$

引理: $[E(XY)]^2 \leq EX^2 \cdot EY^2$

证: $g(t) = E(tX - Y)^2$

$$= E(t^2 X^2 - 2tXY + Y^2)$$

$$= t^2 EX^2 - 2t EXY + EY^2 \geq 0$$

平方期望 ≥ 0 , 为关于 t 的二次函数

$$\Delta = 4[E(XY)]^2 - 4EX^2 \cdot EY^2 \leq 0$$

$$\therefore \Delta = 4[E(XY)]^2 - 4EX^2 EY^2 \leq 0$$

$$\therefore [E(XY)]^2 \leq EX^2 EY^2$$

要证 $|\rho| \leq 1$ 即证 $\rho^2 \leq 1$

$$\text{令 } x_1 = X - EX \quad y_1 = Y - EY$$

$$\rho^2 = \frac{(E[(X-EX)(Y-EY)])^2}{DX DY}$$

$$= \frac{[E(x_1, y_1)]^2}{DX DY}$$

$$\begin{aligned} \because DX &= EX_1^2 & DY &= EY_1^2 \\ &= E(X-EX)^2 & &= E(Y-EY)^2 \end{aligned}$$

$$\rho^2 = \frac{E[(X-EX)(Y-EY)]^2}{E(X-EX)^2 E(Y-EY)^2}$$

$$= \frac{[E(x_1, y_1)]^2}{EX_1^2 EY_1^2}$$

由引理 $[E(x_1, y_1)]^2 \leq EX_1^2 EY_1^2$

$$\therefore \rho^2 \leq 1 \quad \therefore |\rho| \leq 1$$

即 $-1 \leq \rho \leq 1$.

定理: $|\rho| = 1 \Leftrightarrow X$ 与 Y 以 $\rho = 1$ 成线性关系

$$P(Y = \alpha X + b) = 1$$



+++

① $\rho=1$, X, Y 完全正相关.

② $\rho=-1$, X, Y 完全负相关.

③ $|\rho| \rightarrow 0$ 时表示 X, Y 的线性关系弱

④ $\rho=0$ X, Y 不存在线性关系.

X, Y 不相关.

X, Y 独立.

↓
线性不相关

↓
独立 (包括线性非线性在内没有任何关系)

① X, Y 独立, 则 X, Y 不相关

② X, Y 不相关, X, Y 不一定独立
(无线性关系, 但可能有非线性关系)

③ $\because X, Y$ 独立 $\therefore E(XY) = EXEY$.

$$\rho = \frac{E(XY) - E(X)E(Y)}{\sqrt{D_X} \sqrt{D_Y}} = 0 \quad X, Y \text{ 不相关}$$

④ X, Y 不相关, $\rho=0$

$$E(XY) - E(X)E(Y) = 0.$$

$$\text{独立} \Leftrightarrow f(x, y) = f_X(x) f_Y(y)$$

$$\begin{aligned} E(XY) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy f(x, y) dx dy \\ &= \int_{-\infty}^{+\infty} x f(x, y) dx \int_{-\infty}^{+\infty} y f(x, y) dy \end{aligned}$$

例: $Y = X^2$ $\text{cov}(X, Y) = 0$ $\rho = 0$ 不相关.

二维正态分布 (X, Y) 独立与不相关是等价的

4.5 中心矩与原点矩.

原点矩: EX^k 期望: EX

$(X-0)^k$ (-1 阶原点矩)

中心矩: $E(X-EX)^k$.

↘ 以 EX 为中心

-1 阶中心矩: $E(X-EX) = E(X) - EX = 0$

2 阶中心矩: $E(X-EX)^2$ 方差.

① 离散型 $\sum x_i^k p_i$

② 连续型 $\int_{-\infty}^{+\infty} x^k f(x) dx$.

① 离散型 $\sum (x_i - EX)^k p_i$

② 连续型 $\int_{-\infty}^{+\infty} (x - EX)^k f(x) dx$

高于 4 阶的矩极少使用.



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